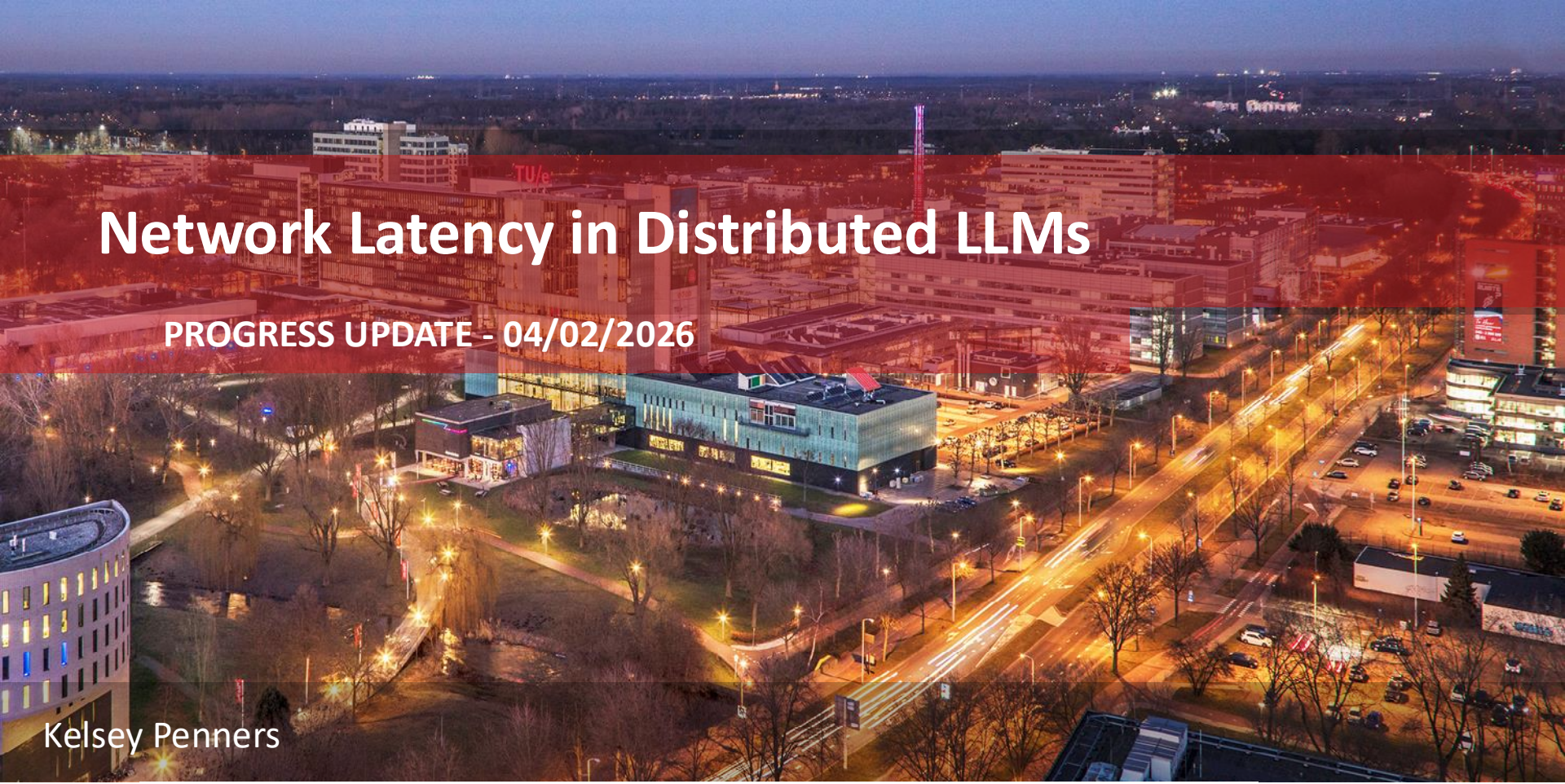


Network Latency in Distributed LLMs

PROGRESS UPDATE - 04/02/2026

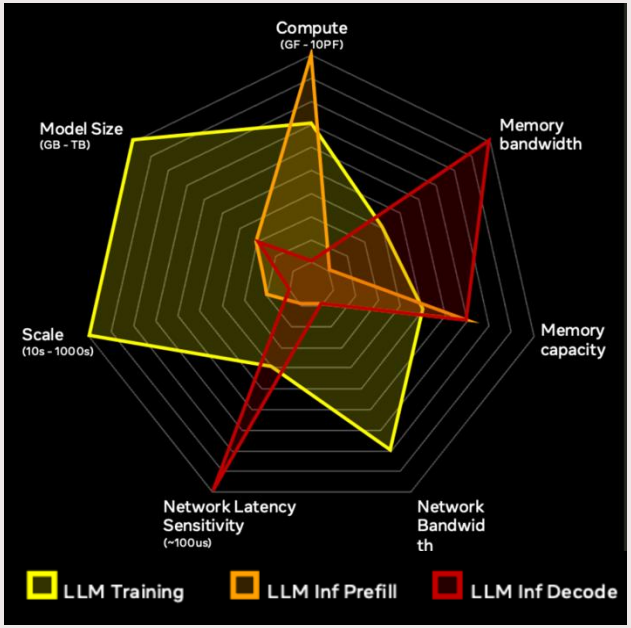
Kelsey Penners

Project Overview

Goal: Characterize how communication latency affects distributed LLM performance (single GPU, multi-GPU, & multi-server)

Progress:

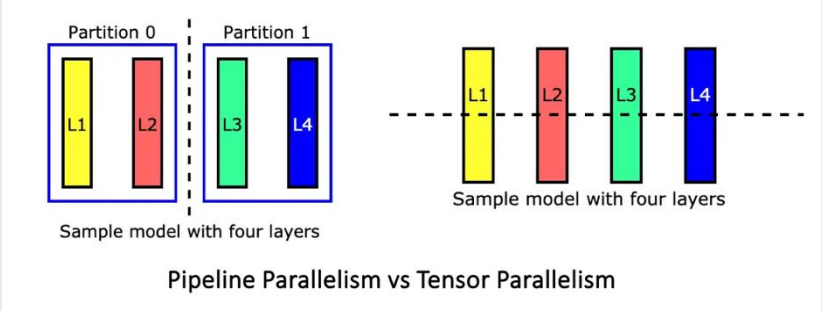
- vLLM setup on shy-fec and documentation exploration, some difficulty setting it up on other servers
- Reading up on relevant papers, documentation, other resources
- Learned accessible server specs, GPU memory, interconnects overview



Source: Ethernet Alliance TEF 2024 Keynote – Ethernet – The foundation of AI @ Meta, Nic Viljoen

Distributed Inference Methods

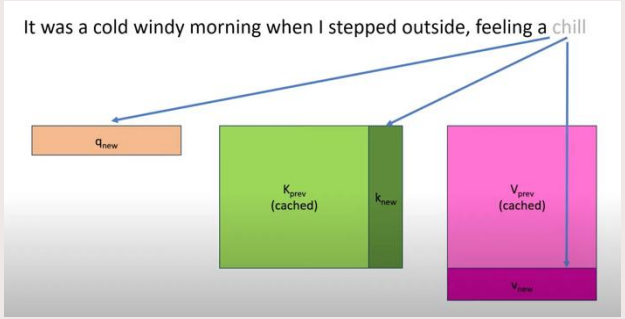
- *Tensor parallelism* (intra-layer / Megatron): splits matrix multiplications across GPUs (within node)
- *Pipeline parallelism*: splits layers across GPUs, (across nodes)
- vLLM supports both, easy enough to handle
- NCCL primitives used to handle GPU communication
  - Routines: allreduce, broadcast, reduce, allgather, reducescatter, P2P
- TP needs high bandwidth (frequent allreduce)
- PP needs low latency (fewer p2p transfers



https://medium.com/tr-labs-ml-engineering-blog/tensor-parallel-llm-inferencing-09138daf0ba7

PagedAttention & vLLM

- KV cache stores key value tensors of the requests
  - Default: stored in contiguous memory, resulting in memory waste
- **PagedAttention** divides KV cache into blocks or pages, stored anywhere in mem, accessed via lookup table (KV block manager)
- Benefits: supports multiple generations, memory-efficient



https://medium.com/my-musings-with-llms/understanding-kv-cache-and-paged-attention-in-llms-a-deep-dive-into-efficient-inference-62fa372432c

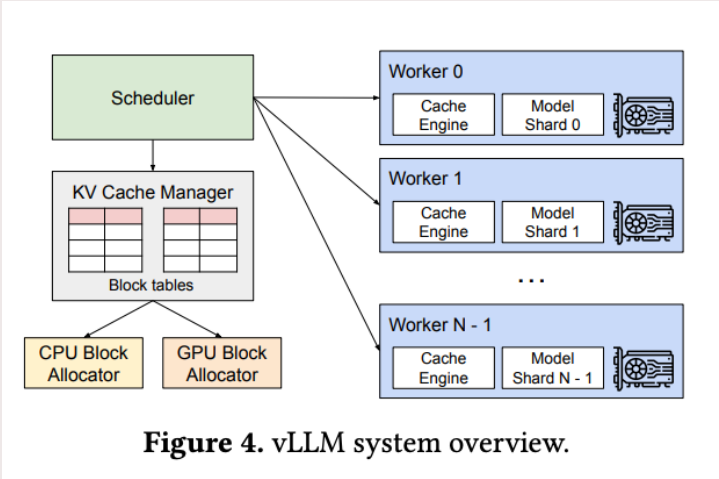


Figure 4. vLLM system overview.

https://arxiv.org/pdf/2309.06180

Servers Overview

Server	GPUs	Total GPU mem	Interconnect & BW	Use?	Largest Model to Fit? (FP16)
spike	32 x B200-180GB (4 DGX systems)	1440 GB per sys, total = 5760 GB	NVLink 1.8 TB/s	Large scale / multi-node	~600b per sys
shy-fec	8 x NVIDIA A100-40GB	320 GB	NVLink 600 GB/s & PCIe 64 GB/s	Primary experiments	~150b
funnotch	4 x Tesla V100-PCIe-32GB	128 GB	PCIe 32 GB/s	PCIe comparison / heterogenous testing	~50b
iconic	4 x Tesla P100-PCIe-16GB	64 GB	PCIe 32 GB/s		~30b
nlcap	4 x Tesla P100-PCIe-16GB	64 GB	PCIe 32 GB/s		~30b

Open Source Models

Provider	Models	Params	Arch.	Does it fit on server?				
				nlcap	iconic	funnotch	shy-fec	spike
Meta (Llama)	**Llama 3.1**	**8b**	Dense	✅	✅	✅	✅	✅
	Llama 3.3	70b	MoE	❌	❌	✅	✅	✅
	Llama 4 Scout	109b	MoE	❌	❌	❌	✅	✅
	Llama 4 Maverick	400b	MoE	❌	❌	❌	✅	✅
DeepSeek	V2 Lite	16b	MoE	❌	❌	✅	✅	✅
	V2	236b		❌	❌	❌	❌	✅
Mistral	**Ministral 3**	**3b**	Dense	✅	✅	✅	✅	✅
		14b	Dense	⚠️	⚠️	✅	✅	✅
	Mistral Large 3	675b	MoE	❌	❌	❌	❌	⚠️
Snowflake Labs	Snowflake Arctic	480b	Hybrid	❌	❌	❌	❌	✅

bolded models = focus for first experiments

✅ fits, ⚠️ fits with restraints (batch/context), ❌ does not fit

TU/e

Next Steps & Such

- Short term: single-GPU dense model baseline on shy-fec, can do TP multi-GPU scaling, maybe PCIe comparison
- Set up project plan: methods, experiments, measurements, models
- Maybe look into disaggregated inference architecture.
- More information on heterogeneous setups

Questions:

- How deep should I go into physical layer, FEC, optics, etc.?
- Best way to reserve GPUs? *CUDA_VISIBLE_DEVICES*?
- nlcap access?

Possible Planned Experiments

- *Baseline:* single GPU, dense small model on shy-fec
  - Measure TTFT, throughput, inter-token latency,
- *Intra-node scaling:* TP1 to TP 8, could compare NVLink vs PCIe (shy-fec vs funnotch), continue with same model and metrics, also try bigger models
- *Inter-node scaling:* use Ray to get nodes working together, run prev models, but also larger ones. Latency injection to happen here
- *Heterogeneous inter-node testing:* test baseline model on different systems